

RepCount Dataset — Description & Analysis

Overview

RepCount is a **repetition action counting dataset** developed by Gao's Lab at ShanghaiTech University. It contains videos with significant variations in length and supports multiple anomaly cases, with fine-grained annotations marking the beginning and end of each action cycle. In total, it provides **1,451 videos** with **19,280 annotations**, and videos average **39.36 seconds** in length.

- **Official Page:** https://svip-lab.github.io/dataset/RepCount_dataset.html
 - **GitHub:** <https://github.com/SvipRepetitionCounting/TransRAC>
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Dataset Structure

The dataset is divided into **two subsets**:

Subset	Source	Split
Part-A	YouTube (1,041 videos)	Train / Validation / Test
Part-B	Recorded volunteers (junior school students & teachers)	Test only

Part-A includes workout activities (squatting, pull-ups, front raising, etc.), athletic events (rowing, pommel horse), and other repetitive actions like soccer juggling. Part-B records exercises such as sit-ups and pull-ups done by volunteers, and is intended specifically to **test model generalization**.

Annotation Files (CSV/TXT Structure)

The dataset ships with annotation files per split: `train.csv`, `valid.csv`, `test.csv`.

Column Descriptions

Column	Name	Description
1	<code>name</code> /	Filename or unique identifier of the video clip

	video_id	
2	url	Original YouTube URL (for Part-A)
3	type / action	Action category label (e.g., squat, pullup, rowing)
4	num_frames	Total number of frames in the video
5	count	Ground truth total number of repetition cycles in the video
6	start_X / end_X	Frame-level timestamps of the start and end of each individual cycle (multiple column pairs, one per repetition)

Note: The start/end columns are dynamic — a video with 10 repetitions will have 10 start/end column pairs. Videos with fewer repetitions will have empty/null values in the remaining columns.

Row Description

Each **row** = one video clip, containing:

- Its metadata (source URL, action type, frame count)
- Its **global count** (total repetitions)
- Its **temporal localization** (where each repetition begins and ends, in frame numbers)

Annotation Types

The labels are composed of two main parts:

1. **Count** — the total number of repetition cycles in the video
2. **Location** — the temporal position of each cycle on the time axis (start and end frame of every individual repetition)

This dual annotation supports two levels of tasks:

- **Counting tasks** — predict how many repetitions occurred
- **Localization tasks** — predict *when* each repetition happened

Key Research Considerations

Aspect	Implication for Research
Variable-length videos	Models must handle temporal scale variation
Anomaly cases	Incomplete or irregular repetitions are included — models must be robust
Two-part structure	Part-B enables cross-domain generalization testing
Fine-grained temporal labels	Enables both counting and segmentation/localization benchmarks

Citation

```
@article{hu2022transrac,  
  title={TransRAC: Encoding Multi-scale Temporal Correlation with Transformers for Repetitive Action Counting},  
  author={Hu, Huazhang and Dong, Sixun and Zhao, Yiqun and Lian, Dongze and Li, Zhengxin and Gao, Shenghua},  
  journal={arXiv preprint arXiv:2204.01018},  
  year={2022}  
}
```

Recommended Next Steps

1. **Download the dataset** and run a statistical analysis on the `count` column to understand the distribution of repetition counts.
2. **Inspect the action type distribution** to assess class imbalance.
3. **Review the TransRAC paper** as the primary baseline method for this dataset.
4. **Compare with related datasets** (e.g., UCFRep, Countix) to contextualize RepCount's contributions.

